INCH-POUND
MIL-DTL-24640/24D
22 August 2002
SUPERSEDING
MIL-C-24640/24C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, MULTI-PAIR, AWG 22, TYPE TTXW (INCLUDING VARIATION TYPE TTXOW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Watertight)

First - Copper conductor, uncoated, AWG 22, 7 strands, in accordance with ASTM B 8.

Second - Mica glass tape insulation 0.0029 ±.0002 inch (50 percent overlap maximum and minimum overlap of 45 percent). Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling.

Third - Opaque-pigmented polyimide tape .001 inch (50 percent overlap minimum).

Fourth - Opaque-pigmented polyimide tape .001 inch with 0.0001 inch Fluorinated Ethylene Propylene (FEP) (50 percent overlap minimum).

Fifth - Modified aromatic polyimide resin coating 0.0005 inch minimum (.001 inch nominal). Identification codes applied using Method 3 (telephone color code) or 6. (Note for method 6, twisted pair identification code colors shall be one black and one white conductor). For type TTXW (OW) 1 1/2, colors shall be one white one black and one red conductor.

Sixth - Two conductors twisted together. (Except for type TTXW (OW) 1 1/2, shall be 3 conductors twisted together to form a triad.) The pitch shall be 1½ ± ¼ inch.

Seventh - The required number of component wires shall be cabled together. Fillers may be used as necessary, to form a round cable.

Eighth - Binder tape applied helically with overlap.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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CONSTRUCTION VARIANT FOR TYPE TTXW

Ninth - Cross-linked polyolefin jacket. (See Table I for thickness).

CONSTRUCTION VARIANT FOR TYPE TTXOW

Ninth - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface transfer Impedance requirements as listed herein.
Tenth - Binder tape applied helically with overlap.
Eleventh –Cross-linked polyolefin jacket. (See Table II for thickness).

TABLE I. Construction details.

PIN M24640/24	Type number	Number of pairs	Jacket thickness (inch) (min)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UN	TTXW-1-1/2	1-1/2	0.028	0.181	0.195	26.0
-02UN	TTXW-3	3	0.038	0.285	0.307	62.3
-03UN	TTXW-5	5	0.037	0.331	0.357	91.0
-04UN	TTXW-10	10	0.041	0.456	0.492	172.0
-05UN	TTXW-15	15	0.044	0.527	0.569	235.0
-06UN	TTXW-20	20	0.044	0.577	0.621	285.0
-07UN	TTXW-30	30	0.044	0.684	0.738	417.0
-08UN	TTXW-40	40	0.047	0.790	0.852	544.0

TABLE II. Construction details - overall shield.

PIN M24640/24	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (min)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UO	TTXOW-1-1/2	1-1/2	0.127	0.031	0.253	0.273	67.9
-02UO	TTXOW-3	3	0.216	0.038	0.333	0.359	108.0
-03UO	TTXOW-5	5	0.263	0.038	0.376	0.406	134.0
-04UO	TTXOW-15	15	0.449	0.044	0.556	0.600	299.0
-05UO	TTXOW-20	20	0.510	0.044	0.614	0.662	371.0
-06UO	TTXOW-30	30	0.610	0.047	0.717	0.772	497.0
-07UO	TTXOW-40	40	0.720	0.047	0.823	0.887	631.0

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REQUIREMENTS:

Qualification required.
Construction details: See Tables I and II.

Finished wire: For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-W-81381:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Blocking: Oven temperature, 200 ± 2°C (392 ± 3.6°F) no sticking of conductors
Flammability: 3 seconds (max) after flame.
3.0 inches (max) flame travel.
No flaming of tissue paper.
Humidity resistance: 5 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test: 100 percent test. Impulse voltage at 8.0 kV.
Insulation resistance: 2500 megohms for 1000 feet (min).
Polyimide cure test: No cracking.
Shrinkage: 0.062 inch maximum at 180°C
Surface resistance: 5 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.031 inch.
Wet dielectric test: 2500 volts (rms).

EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.54
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield (TTXOW only)	1000
Shield to shield	Not applicable
Insulation resistance - megohms-1000 feet, minimum	
Conductor to conductor	2500
Conductor to shield (TTXOW only)	2500
Shield to shield (component shield only)	Not applicable
Shield to shield (component shield to overall shield)	Not applicable
Conductor and shield continuity. (type TTXOW only)	No failure
Jacket flaws (type TTXOW only)	No failure

Group A:
Visual and dimensional.....No failure
Capacitance:

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Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum ..... Not applicable
Capacitance unbalance - percent, maximum ..... Not applicable
Characteristic impedance - at 1 MHz, ohms ..... Not applicable
Watertightness (see Table IX of MIL-DTL-24640 for limits of water leakage) ..... No failure
Weight ..... No failure

Group B:
Cold bending ..... No failure
Drip ..... No failure
Cross link proof test (percent maximum)
Jacket 50
Tear - pounds per inch thickness, minimum 35
Physicals (unaged)
Jacket
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160
Attenuation - at 3 MHz, dB/100 feet, maximum ..... Not applicable
Gas flame (1 hour) ..... No failure

Group C:
Physicals (aged)
Jacket
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged, minimum 60
Permanence of printing (conductor) - Method 6 only cycles, minimum 25
Permanence of printing (jacket) - cycles, minimum 125
Shield - conformance to material, construction and coverage (type TTXOW only) ..... No failure
Heat distortion - percent of unaged, maximum 30
Cable sealant removability ..... No failure
Shrinkage ..... No failure
Mutual inductance - dB/1000 feet, maximum ..... Not applicable
Surface transfer impedance (type TTXOW only)
Milliohms per meter, maximum 700
EMP response, dB, minimum 60
Pulse response time - ms/500 feet
Rise time, maximum ..... Not applicable
Decay time, maximum ..... Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

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Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	No failure
Component wire	No failure

Unit ordering length: 1 1/2 - 15 500 feet nominal.
20 - 40 250 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-024)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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